

Title:

1. FewSOME: One-Class Few Shot Anomaly Detection with Siamese Networks

Method:

- The paper proposes a deep One-Class Anomaly Detection algorithm called **FewSOME** (Few Shot anOMaly dEtectioN).
- It uses a Siamese Network architecture, where two or more neural networks share weights to map input samples into feature embeddings.
- The model is trained on a few normal samples with no anomalous samples.
- The **Stop Loss** is introduced to avoid "Representational Collapse," which happens when all feature embeddings converge to a constant output.

Dataset:

- The paper uses several benchmark datasets for evaluation:
 1. **MNIST**
 2. **CIFAR-10**
 3. **Fashion-MNIST (F-MNIST)**
 4. **MVTec AD** (includes high-resolution images of industrial objects and textures)

Evaluation:

- The performance is evaluated using **AUC (Area Under the Curve)**, **F1 score**, and **Balanced Accuracy**.
- The experiments demonstrate that FewSOME performs at state-of-the-art (SOTA) levels even with a very small number of normal samples, such as 30 normal samples, compared to other methods requiring thousands.
- The robustness to **contaminated datasets** (up to 20% contamination) is also reported.

Technology:

- **Siamese Networks** are used to learn representations of normal data by minimizing the Euclidean distance between normal samples in the feature space.
- The **Stop Loss** prevents representational collapse, which helps the model maintain effective learning even with small datasets.

What the Paper Does:

- The paper introduces FewSOME as a new method for **Few-Shot Anomaly Detection** (FSAD), where the model is trained using only a few examples of the normal class, making it more efficient in real-world applications where anomalies are rare.
- FewSOME is demonstrated to perform well across various anomaly detection benchmarks while significantly reducing training complexity and time.

Experimental Results:

- The ablation study shows the effectiveness of the **Stop Loss** in boosting performance.
- Performance of FewSOME is compared to other methods like **DeepSVDD** and **DROCC**, showing superior results in terms of accuracy and training efficiency.

This paper highlights how FewSOME addresses the challenge of anomaly detection with limited training data and without any examples of anomalies.

2. Paper Name: Time-Series Few Shot Anomaly Detection for HVAC Systems.

- **Method:** The paper introduces a few-shot domain adaptation method based on Long Short-Term Memory (LSTM) autoencoders (AE). This method requires nominal instances from the source domain and a few nominal instances from the target domain. It involves training an anomaly detection classifier on the source domain and fine-tuning it with a small set of nominal instances from the target domain.
- **Dataset:** The dataset used in the study is the MZVAV-2 dataset from the ASHRAE 1312 project, released by Lawrence Berkeley National Laboratory (LBNL) in 2019. It contains real and simulated data from a multi-zone air handling unit in an Iowa building, with both normal operation and faults (e.g., leaking heating coil valve, stuck cooling coil valve).

The dataset consists of time-series data with 17 features and a ground truth label indicating faults or normal operation.

- **Evaluation:** The model is evaluated using performance metrics like True Positive Rate (TPR), False Positive Rate (FPR), and the Area Under the Receiver Operating Characteristic Curve (AUC-ROC). Results show that the proposed method outperforms unsupervised methods like LSTM-AE and clustering-based autoencoders. It achieved the highest AUC of 0.873 and the best TPR of 0.809 when using three nominal instances from the target domain for fine-tuning.
- **Technology:** The method utilizes LSTM autoencoders for anomaly detection and incorporates a few-shot learning technique to reduce the amount of labeled data needed for model adaptation.
- **What the Paper Does:** This paper develops and demonstrates a domain adaptation-based anomaly detection method for HVAC systems. It shows how the LSTM-AE model, when fine-tuned with a few-shot approach, can efficiently detect anomalies in HVAC systems with limited labeled data from the target domain.
- **Experimental List:**
 1. **Data Preprocessing:** Prepares source and target domain datasets, capturing spatio-temporal relationships with LSTM input.
 2. **LSTM-AE Training:** Trains the LSTM-AE model on the source domain's nominal data.
 3. **Fine-Tuning:** Fine-tunes the model on a small number of nominal instances from the target domain.
 4. **Threshold Setting:** Sets a reconstruction error threshold to classify anomalies.
 5. **Anomaly Detection:** Applies the model to test data, classifying anomalies based on the threshold.

This approach effectively reduces the need for large labeled datasets and improves anomaly detection accuracy in real-world HVAC applications.

3. Paper Name: See it, Think it, Sorted: Large Multimodal Models are Few-shot Time Series Anomaly Analyzers

- **Method:** The paper introduces TAMA (Time series Anomaly Multimodal Analyzer), a framework that leverages Large Multimodal Models (LMMs) for anomaly detection in

time series data. The method transforms time series into images for better analysis using LMMs and employs a three-step process: Multimodal Reference Learning, Multimodal Analyzing, and Multi-scaled Self-reflection.

- **Dataset:** The evaluation was performed on several real-world datasets across different domains such as web services (SMD), industry (UCR), health care (ECG), and transportation (Dodgers), as well as synthetic datasets (e.g., NormA, Synthetic).
- **Evaluation:** TAMA outperformed state-of-the-art methods across various metrics, including point-adjusted F1, AUC-PR, and AUC-ROC, showing superior performance, especially on the UCR, NASA-SMAP, and NASA-MSL datasets. The framework's ability to classify anomalies and provide detailed semantic analysis was also highlighted.
- **Technology:** The paper uses Large Multimodal Models (e.g., GPT-4o), which excel in analyzing time series data transformed into images for more accurate anomaly detection and classification.
- **What the Paper Does:** The paper proposes TAMA, a framework that improves time series anomaly detection by converting data into visual formats for better interpretation by LMMs, thus enhancing both detection accuracy and interpretability. It also provides insights into the causes of anomalies through natural language-based explanations.
 - **Experiment List:** The experiments performed include anomaly detection and classification tasks across various datasets, with detailed comparisons to baseline models like LOF, TranAD, and MAD_GAN. Additionally, an ablation study was conducted to test the impact of different hyperparameters like LMM selection, reference number, and window size.

4. Paper Name: FastRecon: Few-shot Industrial Anomaly Detection via Fast Feature Reconstruction

- **Authors:** Zheng Fang, Xiaoyang Wang, Haocheng Li, Jiejie Liu, Qiugui Hu, Jimin Xiao
- **Method:** The paper introduces a few-shot anomaly detection strategy based on feature reconstruction. It uses a few normal samples as a reference to reconstruct the normal version of a defective query sample. The transformation is learned through a regression with distribution regularization, ensuring the reconstructed sample shares visual similarity with the query while maintaining normality.
- **Dataset:**
 - MVTec AD (includes 5354 images in 15 categories)

- MPDD (metal parts dataset with 888 normal and 282 abnormal images)
- **Evaluation:** The method was evaluated on the MVTec AD and MPDD datasets. It outperforms existing state-of-the-art methods in 2-shot, 4-shot, and 8-shot settings, achieving improvements in both image-level and pixel-level AUROC metrics.
- **Technology:** The approach uses feature extraction from a pre-trained ResNet-50 model, followed by a coreset selection to reduce feature redundancy and improve speed. The anomaly detection is performed by aligning the reconstructed sample with the query.
- **What the Paper Does:** The paper proposes FastRecon, a training-free, few-shot anomaly detection approach. It performs well with a limited number of training samples (less than 8) and achieves competitive results compared to methods using large datasets of normal samples.
- **Experiment List:**
 - Feature extraction using ResNet-50
 - Coreset selection to improve speed and reduce data redundancy
 - Regression with distribution regularization to ensure the reconstruction shares normality
 - Comparison with other methods like GANomaly, PatchCore, RegAD, and others
 - Evaluation on MVTec AD and MPDD datasets across various shot settings (2-shot, 4-shot, 8-shot)

The results demonstrate that FastRecon not only generalizes well to different product categories but also achieves high performance with fewer normal samples, which is ideal for real-world industrial anomaly detection.

5. Paper Name: *A Survey of Few-Shot Learning for Biomedical Time Series*

1. **Method:** The paper discusses various few-shot learning methods for biomedical time series, including data-based, model-based, metric-based, optimization-based, and hybrid methods.
2. **Dataset:** It reviews datasets like *PhysioNet*, *MIMIC-IV*, and others relevant to the field of biomedical time series.

3. **Evaluation:** The methods are evaluated using metrics like accuracy (AC), area under the receiver operating characteristic curve (AUC), sensitivity, specificity, and others, with performance improvements compared to traditional learning models.
4. **Technology:** Technologies involved include generative adversarial networks (GANs), convolutional neural networks (CNNs), support vector machines (SVM), and meta-learning frameworks such as Model-Agnostic Meta-Learning (MAML).
5. **What the Paper Does:** The paper provides a comprehensive survey of few-shot learning techniques and their applications in biomedical time series, addressing challenges such as data scarcity and the need for personalized healthcare solutions.
6. **Experiment List:** The experiments reviewed in the paper involve various datasets and applications, such as ECG classification, sleep staging, and seizure detection, applying few-shot learning techniques like GANs, Siamese Networks (SN), Prototypical Networks (PN), and more, across diverse medical tasks.

This survey also highlights emerging trends in few-shot learning, such as multi-modality, foundation models for time series, and the use of neural adaptive processes for model personalization. It also discusses challenges like the lack of standardized benchmarks and the need for better pre-training datasets.

6. Paper Name: Open-Set Multivariate Time-series Anomaly Detection

Method:

- **MOSAD (Multivariate Open-Set time-series Anomaly Detector):** A novel algorithm designed to tackle open-set time-series anomaly detection (TSAD). It leverages a few labeled anomalies during training to detect both seen and unseen anomalies effectively.
- **Key Components:**
 - **Feature Extractor:** Utilizes Temporal Convolutional Networks (TCN) for feature extraction.
 - **Multi-Head Network:** Includes three specialized heads:
 1. **Generative Head:** Models normal data patterns.
 2. **Discriminative Head:** Maximizes the distance between normal and anomaly data.

3. **Anomaly-Aware Contrastive Head:** Uses contrastive learning to better differentiate anomalies.
- **Anomaly Scoring:** Combines scores from the three heads to detect anomalies in both seen and unseen data.

Dataset:

The paper uses three real-world multivariate time-series datasets:

1. **Server Machine Dataset (SMD):** Industrial data from machines in a large internet company.
2. **Physikalisch-Technische Bundesanstalt XL (PTB-XL):** Biomedical ECG data for heart rhythm analysis.
3. **Temple University Hospital Electroencephalogram Seizure Corpus (TUSZ):** EEG data for seizure detection.

Evaluation:

- **Metrics:** Evaluates performance using threshold-independent metrics like **AUC (Area Under the ROC Curve)** and **APR (Area Under the Precision-Recall Curve)**.
- **Experimental Settings:**
 1. **General Setting:** Uses labeled anomalies from all possible anomaly classes.
 2. **Hard Setting:** Uses labeled anomalies from only one anomaly class, making it more challenging.
- **Results:** MOSAD outperforms both unsupervised and supervised methods in anomaly detection across various datasets, demonstrating superior performance in detecting both seen and unseen anomalies.

Technology:

- **Deep Learning:** The method uses a multi-head network with temporal convolutional layers, contrastive learning, and generative models for anomaly detection.

- **Contrastive Learning:** Used for anomaly-aware representation learning.
- **Autoencoder Framework:** The Generative Head uses an autoencoder-based structure to learn the normal data patterns.

What This Paper Does:

- **Addresses Open-Set TSAD:** It tackles the challenge where the system is trained with only a few labeled anomalies and must generalize to new, unseen anomalies.
- **Proposes MOSAD:** A multi-head model that combines generative, discriminative, and contrastive learning approaches to improve anomaly detection performance.
- **Experimental Validation:** The paper demonstrates the effectiveness of MOSAD across real-world datasets, showing that it achieves state-of-the-art results.

Experiment List:

- **SMD Dataset:** Evaluation under unsupervised and open-set settings.
- **PTB-XL and TUSZ Datasets:** Used for open-set and hard settings, testing the model's ability to detect multiple anomaly classes.
- **Ablation Studies:** Different combinations of loss functions and scoring methods are tested to understand the contribution of each component in the overall performance of MOSAD.

7. Paper Name: Few-Shot Anomaly Detection with Adversarial Loss for Robust Feature Representations

- **Method:** The paper proposes a few-shot anomaly detection (FSAD) method that integrates adversarial loss to enhance feature robustness and generalization. The adversarial loss is employed to align feature distributions between the source and target domains, which is particularly beneficial for tasks with limited samples.
- **Dataset:** The experiments were conducted using the MVTec Anomaly Detection (MVTec AD) dataset and the DAGM2007 dataset. The MVTec AD dataset contains images representing normal and defective objects from various categories, while the DAGM2007 dataset includes five classes of images.
- **Evaluation:** The paper evaluates the method using image-level and pixel-level anomaly detection tasks. The performance is compared with existing models like RegAD and UniAD. The proposed method showed improvements in performance across both

datasets and for different shot settings ($K = 2, 4$, and 8).

- **Technology:** The proposed approach utilizes Siamese networks, adversarial training, and discriminator networks to improve feature representations for anomaly detection tasks. The adversarial loss helps generate more generalized and robust features, improving the model's ability to detect anomalies in unseen data.
- **What the Paper is Doing:** The paper aims to enhance anomaly detection performance under few-shot settings by introducing adversarial loss to optimize the use of limited data. It integrates this approach into existing models like RegAD and UniAD to boost their performance.
- **Experiment List:**
 - **MVTec AD dataset:** Evaluates both image-level and pixel-level anomaly detection.
 - **DAGM2007 dataset:** Experiments were conducted to compare the performance of RegAD, UniAD, and the proposed method.

This summary reflects the primary aspects of the paper and its contributions to the field of anomaly detection under few-shot conditions.

8. Paper Name: Dive into Time-Series Anomaly Detection: A Decade Review

Methods:

The paper categorizes time-series anomaly detection methods into three major groups:

1. **Distance-based methods:** These include proximity-based methods (e.g., KNN), clustering-based methods (e.g., k-means), and discord-based methods (e.g., matrix profile).
2. **Density-based methods:** These include distribution-based, graph-based, tree-based, and encoding-based methods (e.g., One-Class SVM, Isolation Forest).
3. **Prediction-based methods:** These methods predict future time-series values and identify anomalies based on prediction errors (e.g., ARIMA, LSTM).

Datasets:

The paper reviews various datasets used for anomaly detection, including:

- **NAB**: 58 time series, used for real-time anomaly detection.
- **Yahoo**: A collection of real and synthetic datasets for anomaly detection.
- **Exathlon**: High-dimensional time series for explainable anomaly detection.
- **KDD21**: Composite dataset covering diverse domains.
- **TSB-UAD**: A comprehensive benchmark for univariate time-series anomaly detection.

Evaluation:

Different evaluation measures include:

1. **Threshold-based evaluation**: Involves setting a threshold to classify data points as anomalies based on their anomaly score.
2. **Threshold-independent evaluation**: Metrics like AUC, Range-AUC, and Volume Under the Surface (VUS) that do not rely on pre-set thresholds.

Technology:

Various technologies discussed in the paper include machine learning methods like **LSTM**, **Autoencoders**, **Generative Adversarial Networks (GANs)**, and traditional statistical methods like **ARIMA**.

Experiments:

The paper evaluates several anomaly detection methods on real and synthetic datasets. It also highlights benchmarks like **NAB** and **Yahoo**, and discusses the challenges and developments in the field over the past decade.

This review helps to understand the growing interest in time-series anomaly detection, especially with the advent of deep learning techniques like LSTM and Autoencoders since 2020.

9.The paper titled "DACAD: Domain Adaptation Contrastive Learning for Anomaly Detection in Multivariate Time Series"

Method:

The DACAD model integrates contrastive learning (CL) with domain adaptation techniques for anomaly detection in multivariate time series (MTS). The model uses:

- **Anomaly Injection:** Synthetic anomalies are introduced to enhance the model's robustness.
- **Supervised Contrastive Loss:** Applied in the source domain using labeled data for anomaly and normal samples.
- **Self-supervised Contrastive Triplet Loss:** Used for the target domain, where labels are absent, to ensure proper feature representation.
- **Center-based Entropy Classifier (CEC):** A classifier that refines anomaly detection by pulling normal sample representations closer to a center point and pushing anomalous ones farther.

Datasets:

DACAD was evaluated on several real-world and synthetic datasets:

1. **MSL (Mars Science Laboratory):** Spacecraft anomaly detection.
2. **SMAP (Soil Moisture Active Passive):** Another space-based anomaly detection dataset.
3. **SMD (Server Machine Dataset):** Anomaly detection from server data.
4. **Boiler Fault Detection Dataset:** Focuses on detecting faults in boilers.

Evaluation:

The model was evaluated using multiple metrics including F1 score, AUPR (Area Under Precision-Recall Curve), and AUROC (Area Under Receiver Operating Characteristic Curve). DACAD consistently outperformed other models, including:

- **VRADA** (Variational Recurrent Adversarial Deep Domain Adaptation).
- **CLUDA** (Contrastive Learning for Unsupervised Domain Adaptation).

Technology:

The model utilizes deep learning, specifically employing temporal convolutional networks (TCNs) to capture temporal dependencies in time series data. It also integrates domain adaptation and contrastive learning techniques.

Experiments:

The DACAD approach was tested using the following experimental strategies:

- **Baseline comparison:** Results were compared to several state-of-the-art TSAD models like OmniAnomaly, THOC, TS2Vec, AnomalyTransformer, and DCdetector.
- **Scalability tests:** DACAD demonstrated superior scalability in terms of execution time and memory usage, particularly when handling larger time series data.

The findings suggest that DACAD is effective in adapting to new, unlabeled domains and performs well even with scarce labeled data, making it a valuable contribution to TSAD.

10.Paper Name: "CAN LLMS UNDERSTAND TIME SERIES ANOMALIES?" by Zihao Zhou and Rose Yu.

Method:

The paper evaluates LLMs' anomaly detection capabilities by testing multiple hypotheses about their performance on time series anomaly detection. It involves experiments where LLMs process time series data in text and visual formats to identify various anomaly types. The study uses different prompting techniques like zero-shot and few-shot learning (FSL), Chain-of-Thought (CoT) reasoning, and other strategies such as Token-per-Digit (TPD) and Dyscalculia-based impairment (DysCalc).

Dataset:

Four main datasets are synthesized, each focused on different anomaly types:

1. Point Anomalies
2. Range Anomalies
3. Trend Anomalies
4. Frequency Anomalies

In addition, noisy versions of the datasets were created to test specific hypotheses about LLMs' ability to handle noisy data. Real-world datasets like Yahoo S5 were also used for validation.

Evaluation:

The paper primarily uses metrics such as Precision, Recall, F1-Score, and Affinity F1 (a more appropriate metric for time series). The performance of various LLM architectures—such as GPT-4o-mini, Gemini-1.5-Flash, Qwen-VL-Chat, and InternVL2—was assessed across different prompting methods.

Technology:

The models evaluated include state-of-the-art multimodal LLMs (M-LLMs) that can process both text and visual inputs, such as Qwen-VL-Chat, InternVL2-Llama3, Gemini-1.5-Flash, and GPT-4o-mini. These models were tested using controlled experiments, with a focus on how LLMs process visual versus text inputs in time series analysis.

What the Paper Does:

The study challenges assumptions about LLMs' anomaly detection capabilities in time series analysis, revealing that while LLMs can detect basic anomalies (e.g., point anomalies), their performance drops significantly with more complex anomaly types. The research shows that LLMs perform better with visual representations of time series data compared to text alone, and that explicit reasoning (CoT) does not necessarily improve performance.

Experiment List:

1. Zero-shot and Few-shot Anomaly Detection
2. Chain-of-Thought (CoT) Reasoning Experiment
3. Testing the Impact of Token-per-Digit (TPD) on Arithmetic Bias
4. Testing Visual versus Textual Data Processing
5. Comparing Performance Across Different LLM Architectures